

Problem Q

Eva's Hair Transplant Clinic

Time limit: 3 seconds

Memory limit: 1024 megabytes

Problem Description

Eva has been staying up late far too often, and recently she noticed she's been losing a lot of hair! Although she felt upset, she also saw a business opportunity. She decided to open a hair transplant clinic!

Because Eva is very busy, she doesn't have time to diagnose each customer in person. So she wants to develop a program, customers only need to upload a photo of the top of their head, and Eva will be able to determine how many hairs they need transplanted.

To do this, Eva wrote a program that converts the uploaded image into a hair-density matrix. Areas without hair are marked as '0', and non-zero numbers indicate denser hair. However, a full image takes up too much memory, so Eva hopes to compress the matrix. The compression method records only the non-zero entries as `'[row, column, value]'`. When Eva needs to make a diagnosis, she needs to reconstruct the full hair-density map from the compressed data.

For example:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 9 \end{bmatrix}$$

and

$$\begin{bmatrix} 3 & 3 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

But as mentioned earlier, Eva is too busy, so she hopes to entrust this task to some brilliant juniors to help her complete this feature.

Input Format

Your program is to read from standard input. The input consists of several test cases. Each test case, the first contains only one integer number C , where '1' indicate the original matrix and '1' indicate the compressed matrix. The second line contains three integers n , m and k , representing the size of the matrix is $n \times m$, and the number of non-zero values is k . If C is 1, each of the next n lines contains m elements, which represents the original $n \times m$ matrix. If C is 2, then each of the next k lines contains

three elements representing the compressed matrix which records only the non-zero entries as '[row, column, value]' format. The program ends when C is '0'.

Output Format

Your program is to write to standard output. For the test case that C is '1', output the matrix after compressed. For the test case that C is '2', output the matrix before compressed. Two consecutive test cases are separated by a blank line. Please see the sample output.

Technical Specification

- $C \in \{1, 2\}$
- $3 \leq n, m \leq 100$
- $1 \leq k \leq n \times m$
- For $C = 1$, the elements in $n \times m$ matrix are integers between 0 and 9.
- For $C = 2$, $1 \leq row \leq n, 1 \leq column \leq m, 1 \leq value \leq 9$.

Sample Input 1

```
1
3 3 1
0 0 0
0 9 0
0 0 0
2
3 3 1
3 3 5
0
```

Sample Output 1

```
2 2 9

0 0 0
0 0 0
0 0 5
```

Sample Input 2

```
1
5 5 5
0 0 0 0 3
5 0 5 0 0
0 0 0 0 0
0 1 0 0 0
2 0 0 0 0
2
5 5 5
1 1 2
2 5 2
4 1 1
4 3 9
```

Sample Output 2

```
1 5 3
2 1 5
2 3 5
4 2 1
5 1 2

2 0 0 0 0
0 0 0 0 2
0 0 0 0 0
1 0 9 2 0
0 0 0 0 0
```

4 4 2
0

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